

PROJECT REPORT

Portfolio Constraint Optimization

Assignment ID: FF26A1

Date: March 07, 2026

1. INTRODUCTION

This project implements a comprehensive portfolio constraint optimization system that applies systematic filters to investment strategies. The objective is to enhance portfolio quality by removing underperforming stocks and ensuring proper diversification across sectors and market capitalizations.

2. METHODOLOGY

2.1 Data Processing

The system processes three key datasets:

Dataset	Records	Purpose
Investment Rules	100 strategies	26 years of portfolio holdings
Sector Data	6,709 companies	Sector classifications
Market Cap Data	7,176 records	Year-wise market capitalizations

2.2 Constraint Application

Constraint 1: Market Capitalization Filter

- Objective: Remove illiquid, low-quality stocks
- Implementation: Rank stocks by market cap, remove bottom 20%
- Rationale: Ensures minimum liquidity and company size thresholds
- Impact: Average 20-25% portfolio reduction per strategy

Constraint 2: Sector Concentration Limit

- Objective: Prevent over-concentration in single sectors
- Implementation: Limit each sector to maximum 25% of portfolio
- Enforcement: Remove lowest MCap stocks from over-concentrated sectors
- Impact: Typically 0-10% additional exclusions

2.3 Algorithm Design

The system implements a two-stage filtering process:

1. **Load and Parse:** Read CSV files, build lookup dictionaries for O(1) access
2. **MCap Filter:** Sort stocks by market cap descending, keep top 80%
3. **Sector Check:** Calculate sector distribution, identify violations
4. **Sector Correction:** For over-concentrated sectors, keep highest MCap stocks
5. **Output Generation:** Write constrained portfolios to output.csv

3. RESULTS

3.1 Quantitative Outcomes

Metric	Value
Strategies Processed	100
Total Stocks Excluded (MCap)	~31,000
Total Stocks Excluded (Sector)	~5,000
Average Exclusion Rate	20-25%
Output File Size	1.90 MB
Processing Time	<60 seconds

3.2 Example Analysis (Strategy 1, Year 2021)

- Original stocks: 158
- After MCap filter: 126 (32 excluded)
- After sector constraint: 126 (0 additional exclusions)
- Final portfolio: 126 stocks across 25+ sectors
- Top sector concentration: Steel (12.7%) - well below 25% limit
- Market cap distribution: 66% mid-cap, 18% large-cap, 17% small-cap

3.3 AI-Generated Insights

The system integrates with AI APIs (OpenAI, Claude, Gemini) to generate professional portfolio analysis reports covering sector analysis, risk assessment, and actionable recommendations. Sample analysis demonstrates the portfolio's industrial bias, mid-cap orientation, and cyclical exposure characteristics.

4. IMPLEMENTATION

4.1 Technology Stack

- Language: Python 3.13
- Data Processing: Pandas (efficient DataFrame operations)
- Web Interface: Streamlit (interactive UI)
- Visualization: Plotly (interactive charts)
- AI Integration: OpenAI, Anthropic, Google Gemini APIs

4.2 Code Structure

- **portfolio_constraint.py**: Core constraint logic (220 lines, fully documented)
- **portfolio_analyzer.py**: Extended functionality with AI integration (400+ lines)
- **app.py**: Streamlit web application (500+ lines)
- Modular design with clear separation of concerns

4.3 Key Features

- ✓ Handles 100 strategies × 26 years = 2,600 portfolios
- ✓ Processes 6,709 companies with sector classifications
- ✓ Dynamic column detection (handles CSV format variations)
- ✓ Robust error handling and logging
- ✓ Interactive visualizations (pie charts, bar graphs)
- ✓ Real-time AI-powered portfolio analysis
- ✓ Downloadable reports in CSV and PDF formats

5. CONCLUSION

This project successfully implements a systematic portfolio constraint optimization system that enhances portfolio quality through MCap filtering and sector diversification enforcement. The solution processes 100 investment strategies efficiently, excluding ~36,000 individual stock positions that fail quality or diversification criteria.

The integration with AI services enables automated generation of professional-grade portfolio analysis reports, providing actionable insights on risk, sector exposure, and optimization opportunities. The Streamlit web application makes the system accessible through an intuitive interface with real-time analysis and interactive visualizations.

Key Achievements:

- Successful constraint application across 2,600 portfolio-year combinations
- Average 20-25% portfolio optimization through systematic filtering
- Comprehensive AI-powered analysis pipeline
- Production-ready web application with complete documentation
- Scalable, maintainable codebase with extensive error handling